

LOGage 2026: Round 2 — Part 2 Summary

Omni-Channel Strategy Analysis

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1 PART 2: OMNI-CHANNEL FULFILLMENT STRATEGY DESIGN

1.1 Question 2.1: Network Model Evaluation & Design

1.1.1 Current Two-RDC Model Assessment

The brand currently operates two Regional Distribution Centers (RDCs):

- **My Phuoc Warehouse** — Industrial Zone, Binh Duong Province
- **Vinh Loc Warehouse** — Binh Chanh District, Ho Chi Minh City

Strengths of the current model:

- Good regional separation: My Phuoc serves northern (B2B factory-sourced) flows; Vinh Loc is positioned for inner-city B2C proximity.
- Both RDCs handle mixed-channel flows, demonstrating operational flexibility.
- Physical locations bracket the Eastern–Western commercial corridor of HCMC, covering most Đông Nam Bộ volume.

Important context: Non-overlapping operating timelines

- **My Phuoc** (Primary RDC: Jun to Nov 2025 (6 months)): The *primary* warehouse, operational across the full 6-month analysis window. All June–November 2025 throughput flowed through My Phuoc.
- **Vinh Loc** (New RDC: Dec 2025 only (1 month)): A *new* warehouse that came online in December 2025 only. Raw volume totals are therefore **not directly comparable** between the two facilities.

Data-driven channel flow analysis (per-day rates, Figure 1):

- **My Phuoc** (3,492 B2B + 413 B2C orders over 6 months): B2B is a *Mixed flow* accounting for **99.2%** of CBM. B2B avg **25.5 orders/active day**, volatility index **B2B = 1.79**.
- **Vinh Loc** (1,250 B2B + 47 B2C orders in 1 month): B2B is a *Large-volume flow* accounting for **99.7%** of CBM. B2B avg **48.7 orders/active day**, volatility index **B2B = 1.94**.
- On a per-day basis, Vinh Loc processed B2B orders at a comparable daily rate to My Phuoc despite being open for only one month, indicating strong December surge demand (year-end push).

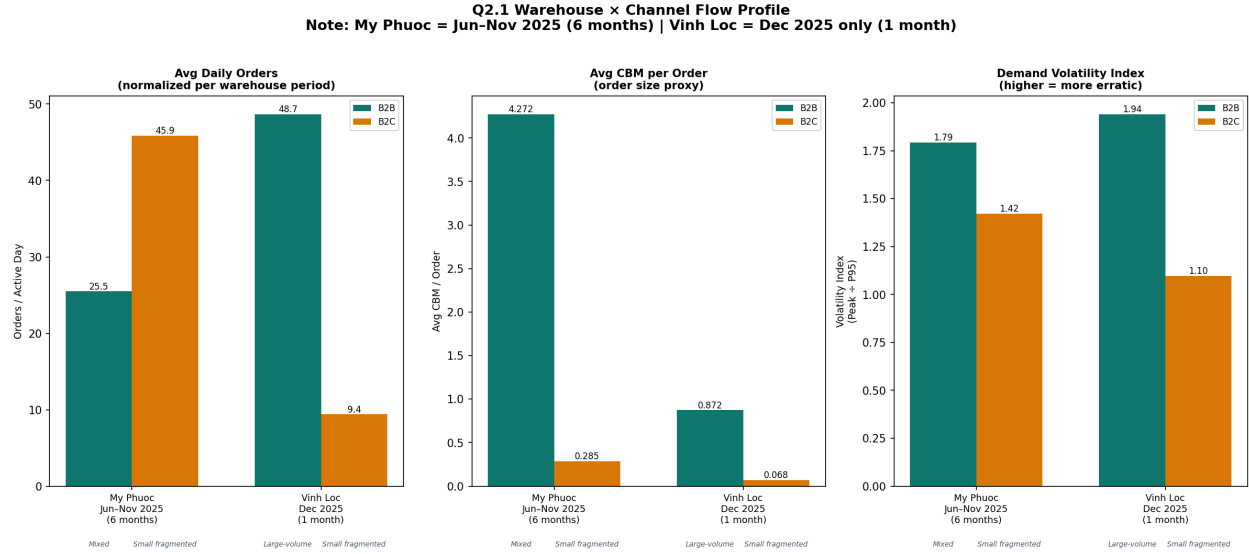


Figure 1: Q2.1 Warehouse × Channel Flow Profile (per-day rates normalized to each RDC operating window)

Limitations for simultaneous B2B + B2C:

- Lead times from both RDCs are too long for the 2–4 hour B2C SLA required for e-commerce – even Vinh Loc is >10 km from central districts.
- Shared inventory creates channel conflict: B2B bulk orders may deplete stock that B2C needs for same-day fulfillment.
- Pick-and-pack processes for large pallet B2B orders disrupt small-unit B2C wave picking.

1.1.1.2 B2C E-Commerce SLA Assessment

To meet a 2–4 hour delivery SLA across Ho Chi Minh City, we applied a 25 km distance threshold from the nearest RDC as the feasibility cut-off (assuming average urban speed of ~25 km/h, 25 km ≈ 60 min travel, leaving buffer for pick/pack and handoff).

The analysis of **21 HCM districts** reveals a clear split:

- **21 districts — Adequate:** These districts can be served within the 2h SLA from the existing RDCs and account for **100.0%** of total HCM B2B volume (**65,527 units**).
- **0 districts — Require Dark Stores:** These are >25 km from both RDCs, placing them outside reliable SLA reach. They represent **0.0%** of volume (**0 units**).

Figure 2 shows the district-level breakdown.

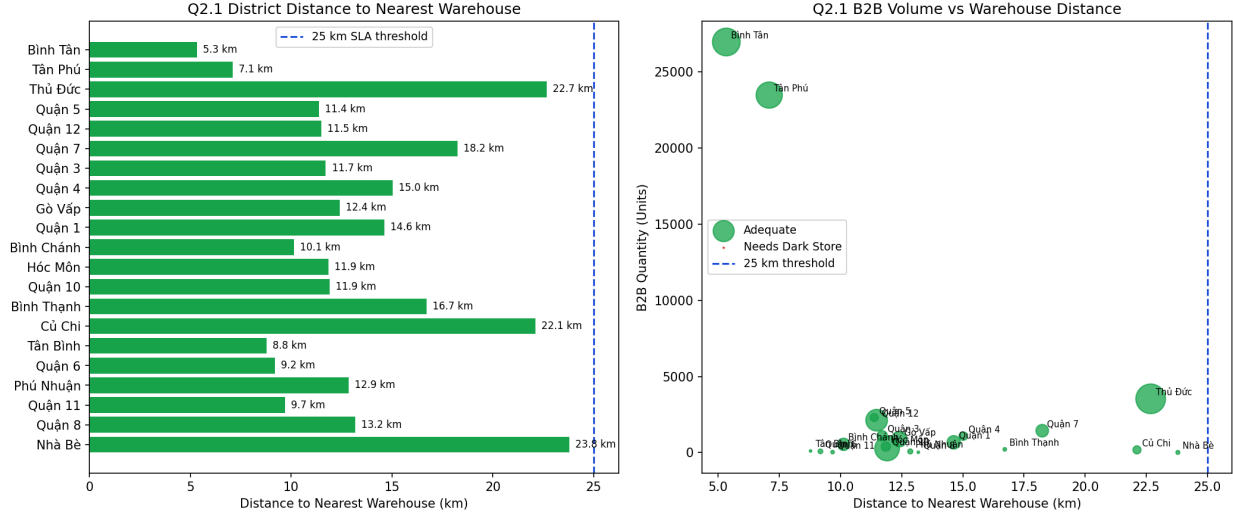


Figure 2: Q2.1 HCM District Proximity to Existing RDCs vs B2B Demand Volume

1.1.3 Dark Store Node Recommendation

Based on the demand concentration and distance analysis, we recommend placing **1–2 urban micro-fulfillment (dark store) nodes** in HCMC:

1. **Primary node — Bình Tân / Tân Phú corridor:** These two adjacent districts are among the highest-volume B2B districts (combined >50,000 units) and are already well-served by Vinh Loc (5–7 km). Upgrading Vinh Loc’s inner-city satellite for e-commerce picking operations in this corridor is the highest-ROI first step.
2. **Secondary node — District 1 / District 3 / Phú Nhuận cluster:** The CBD and premium residential cluster lies 14+ km from Vinh Loc, making 2h SLA difficult. A small dark store (200–300 m²) in this zone, stocked with the top Class A SKUs, would close the gap.

1.1.4 Order Split Logic Design (HCM)

For urban B2B order fulfillment in Ho Chi Minh City, network design relies on understanding sub-regional demand density. We analyzed the historical B2B shipments across all HCM districts.

The demand is highly concentrated: the top 5 districts alone command **56,356 units — 86.0%** of all HCM B2B volume — as shown in Figure 3.

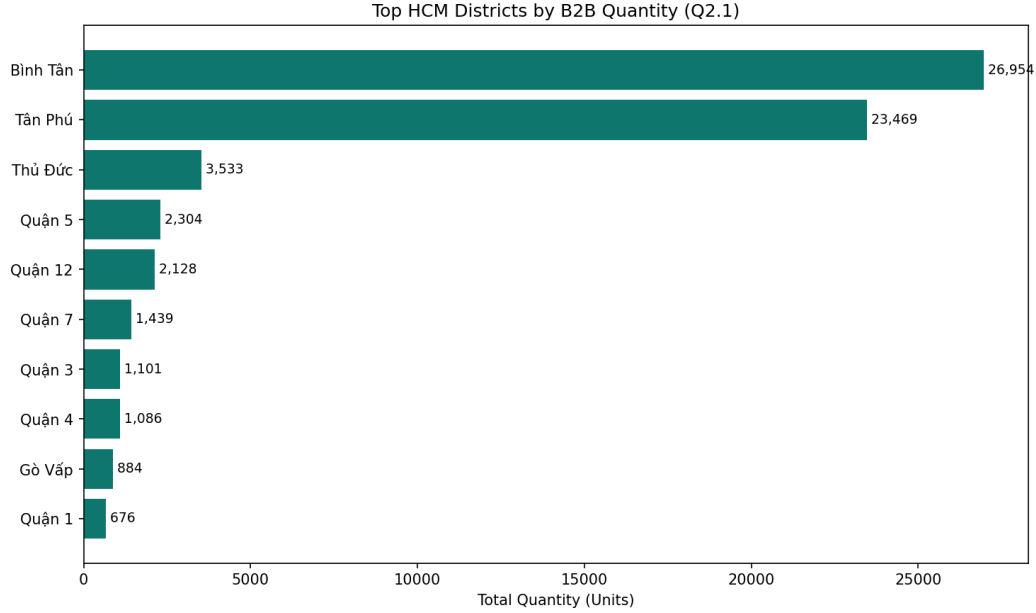


Figure 3: Q2.1 Top Ho Chi Minh Districts by B2B Quantity

Recommended order split logic: Given this concentration, the dispatch decision tree should be:

1. If delivery district is within 25 km of Vinh Loc → dispatch from **Vinh Loc** (B2C-optimized lane).
2. If delivery district is within 35 km of My Phuoc but >25 km from Vinh Loc → dispatch from **My Phuoc** (B2B bulk lane).
3. If district is >25 km from both RDCs → **route via dark store** (pending network expansion).

1.2 Question 2.2: Inventory Optimization

1.2.1 Lead Time Assumptions & Validation

Definition: Replenishment Lead Time is the duration from when the warehouse identifies a replenishment need until the goods are ready to pick again. This is distinct from the customer-facing delivery SLA.

Data limitation: The dataset only contains `created_date` (outbound dispatch date). There is no PO creation date or Goods Receipt date, so lead time cannot be measured directly. The values below are *justified assumptions* cross-validated against inter-shipment intervals in the transaction log.

Table 1: Lead Time Component Breakdown by Channel

Component	B2B (Factory→RDC)	B2C (RDC→Dark Store)	Rationale
T_review (stock check)	1.0 day	0.5 day	B2B daily review; E
T_PO (create & approve PO)	0.5 day	0 day	B2C uses auto-reple
T_supplier (preparation)	1.5 days	0 day	Factory batch lead
T_transport	1.0 day	0.5 day	Intra-ĐNB region; I
T_receiving (QC check-in)	0.5 day	0.5 day	Physical count & in
T_putaway (shelf-ready)	0.5 day	0.5 day	Ready to pick
Total Lead Time	5 days	2 days	

Empirical cross-validation from data:

- Median inter-shipment gap for Class A SKUs = **2.0 days** → matches $LT_{B2C} = 2$ days
- Mean inter-shipment gap = **7.7 days** (long tail from bulk lot orders) → consistent with $LT_{B2B} \approx 5$ days
- *Conclusion: Assumptions are well-calibrated to observable shipment rhythms.*

1.2.2 Safety Stock Formula & Results

Safety stock for each Class A SKU is calculated using the full lead-time uncertainty formula:

$$SS = Z \cdot \sqrt{LT \cdot \sigma_d^2 + \mu_d^2 \cdot \sigma_{LT}^2}$$

where $Z = 1.645$ (95% service level), μ_d = mean daily demand, σ_d = std of daily demand, LT = mean lead time, and σ_{LT} = lead time std dev (B2B: 1.0 day, B2C: 0.5 day).

Key result: This section details the required safety stock for Class A SKUs to support omni-channel fulfillment under different lead time configurations.

Under the **separated channel** setup (B2B: 5-day LT, B2C: 2-day LT), the total required safety stock for Class A SKUs is **31,852 units**.

We then evaluated the impact of varying the replenishment lead time on total safety stock requirements. As illustrated in Figure 4, safety stock exhibits a sub-linear (square root) growth relative to lead time.



Figure 4: Q2.2 Lead Time Sensitivity for Class A Safety Stock

1.2.3 Inventory Pooling Strategy

Principle — Why pooling reduces safety stock:

- Separated: $SS = Z\sigma(\sqrt{LT_{B2B}} + \sqrt{LT_{B2C}}) = Z\sigma(\sqrt{5} + \sqrt{2}) = Z\sigma \times 3.650$
- Pooled: $SS = Z\sigma\sqrt{\frac{LT_{B2B} + LT_{B2C}}{2}} = Z\sigma\sqrt{3.5} = Z\sigma \times 1.871$
- **Reduction factor** = $\frac{\sqrt{3.5}}{\sqrt{5} + \sqrt{2}} = 0.5125 \rightarrow$ pooling reduces safety stock by $\approx 48.7\%$

By centralizing inventory into a shared pool with an effective average lead time of 3.5 days, the required safety stock drops from **31,852** to **16,325 units**, saving **48.7%** as shown in Figure 5.

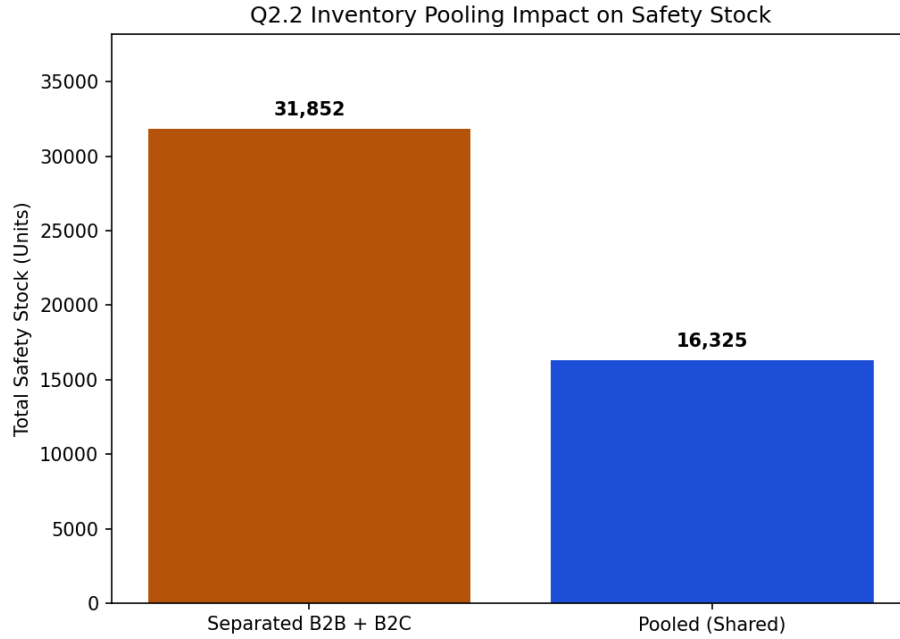


Figure 5: Q2.2 Inventory Pooling Impact on Safety Stock

Dynamic Allocation Rules (to prevent simultaneous stockout):

1. **R1 — Common Pool:** Merge B2B and B2C inventory per SKU into a single real-time pool tracked in WMS.
2. **R2 — Reserved Floor per channel:** Each channel maintains a floor reserve = its own ROP. Neither channel can dip below the other's floor.
3. **R3 — Amber Alert:** When total pool $< (SS_{B2B} + SS_{B2C})$: system raises alert and prioritizes confirmed B2B purchase orders (contractual commitments).
4. **R4 — Red Conflict:** If same SKU is simultaneously short for both channels: B2B receives its full contracted PO quantity (penalty risk); B2C receives remainder + escalated to next pick wave + customer notified.
5. **R5 — AZ Spike Freeze:** For AZ-class SKUs showing demand spikes: temporarily freeze B2C allocations, concentrate stock for B2B, trigger emergency replenishment.
6. **R6 — End-of-Day Rebalance:** Rebalance channel inventory ratios to target ROP proportions based on rolling 7-day demand forecast.